

AN INTERNSHIP REPORT

Of

Python Programming Virtual Internship

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

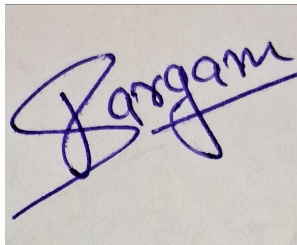
Name: Sargam Yadav
Roll Number: 25SCS1003003247
Batch: 2025–29

CANDIDATE'S DECLARATION

I, Sargam Yadav, do hereby declare that the internship report titled "Python Programming" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The work presented in this report is based on the projects completed during the Python Programming virtual internship with CODSOFT. All references used for the preparation of this report have been acknowledged appropriately.

The report describes three practical Python applications developed during the internship: Rock Paper Scissors, Calculator, and Password Generator. The report is prepared for academic submission and has not been submitted to any other institute for the same degree or diploma requirement.

Signature

A handwritten signature in blue ink, appearing to read 'Sargam', is written on a light-colored background. The signature is stylized with a large initial 'S' and a horizontal line extending from the end.

Student Name - Sargam Yadav
Roll No. - 25SCS1003003247

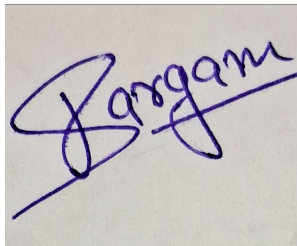
ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for the guidance, constructive feedback, and encouragement received during the completion of the assigned projects.

The internship provided valuable exposure to practical Python programming, application development, graphical user interfaces, testing, debugging, and project organization. The experience helped in understanding how programming concepts can be applied to develop simple and functional applications.

I also express my gratitude to my parents for their blessings and continuous support throughout this learning experience.

Signature

A handwritten signature in purple ink, appearing to read 'Sargam', is shown on a light-colored background.

Student Name - Sargam Yadav
Roll No. - 25SCS1003003247

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3. INTERNSHIP COMPLETION CERTIFICATE



The certificate confirms successful completion of the Python Programming virtual internship at CODSOFT from 05 August 2026 to 05 September 2026.

4. INTERNSHIP OFFER LETTER



INTERNSHIP OFFER LETTER

Date : 02/08/2026

ID: BY26RY227375

Dear **Sargam Yadav** ,

We would like to congratulate you on being selected for the “**Python Programming**” virtual internship position with “**CodSoft**”. We at CodSoft are excited that you will join our team. The duration of the internship will be of **1 Month**, starting from **05 August 2026 to 05 September 2026**. The internship is an educational opportunity for you hence the primary focus is on learning and developing new skills and gaining hands-on knowledge. We believe that you will perform all your tasks/projects. As an intern, we expect you to perform all assigned tasks to the best of your ability and follow any lawful and reasonable instructions provided to you.

We are confident that this virtual internship will be a valuable experience for you, we look forward to working with you and helping you achieve your career goals.

By accepting this offer, you commit to executing assigned tasks diligently and ensuring excellence in all aspects of your work.

Best of Luck!

Sincerely

A handwritten signature in black ink that reads "Ruman Arshad".

Ruman Arshad
Program Manager



ISO Number - TSNUK79053



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5. PROJECT DESCRIPTION

5.1 Introduction

Python is a high-level programming language widely used for application development because of its readable syntax and wide range of libraries and modules. It supports rapid development and is suitable for creating both basic and advanced applications.

The Python Programming Internship involved the development of three practical applications: Rock Paper Scissors, Password Generator, and Calculator. Each project was developed to apply fundamental Python concepts in a practical environment.

The projects provided hands-on experience with programming logic, functions, user input, conditional statements, randomization, operators, modules, graphical user interfaces, testing, and debugging. Tkinter was used to create interactive graphical interfaces for the applications.

The overall project work demonstrates how basic programming concepts can be combined to develop useful and functional applications.

5.2 Organization Profile

CODSOFT is an organization that provides internship opportunities focused on practical learning and skill development. The internship program provides students with project-based tasks that allow programming concepts to be applied to practical applications.

The Python Programming Internship focused on developing applications using Python and understanding the basic process of application development. The assigned projects provided practical exposure to coding, interface development, testing, debugging, and project organization.

The major projects completed during the internship were:

- Rock Paper Scissors
- Calculator
- Password Generator

5.3 Problem Statement

Rock Paper Scissors: A traditional hand game can be converted into a digital application where a user plays against the computer. The application needs to accept the player's choice, generate a computer choice, compare both choices, and display the result.

Calculator: Basic mathematical operations are frequently required in academic and everyday activities. An interactive calculator can provide a quick and convenient way to perform common mathematical calculations.

Password Generator: Creating passwords manually may result in repetitive or predictable combinations. A password generation tool can simplify the process by producing random passwords using different characters according to the required length.

5.4 Project Objectives

The major objectives of the internship projects were:

- To develop practical applications using Python.
- To strengthen understanding of Python programming fundamentals.
- To improve logical thinking and problem-solving skills.
- To understand the use of functions, operators, lists, strings, and modules.
- To use randomization for game and password-generation logic.
- To develop interactive graphical interfaces using Tkinter.
- To implement user input and event handling.

Project-Specific Objectives

Rock Paper Scissors: To provide a simple interactive game in which the user competes against a computer-generated choice.

Calculator: To perform common arithmetic operations through an interactive interface.

Password Generator: To generate random passwords based on user-defined requirements.

5.5 Scope of the Project

The developed applications provide basic functionality and can be further improved with additional features.

Rock Paper Scissors: The application can be extended with score tracking, multiple rounds, difficulty levels, sound effects, player statistics, and different game modes.

Calculator: The calculator can be expanded with scientific calculations, calculation history, memory functions, keyboard support, and advanced mathematical operations.

Password Generator: Future improvements may include a password strength indicator, copy-to-clipboard functionality, password history, and additional customization options.

The current projects provide a foundation for developing more advanced applications in the future.

5.6 Technologies and Tools Used

Python: Python was used as the primary programming language for developing all three applications. It was used to implement application logic, functions, calculations, user input, randomization, and required processing.

Tkinter: Tkinter was used to design the graphical user interfaces of the applications. It provided interactive components such as windows, buttons, labels, and input fields.

Random Module: The random module was used where random choices were required, especially for computer moves in Rock Paper Scissors and for generating password characters.

String Module: The string module was used to access standard character sets for password generation.

Visual Studio Code: Visual Studio Code was used as the development environment for writing, executing, and debugging the Python programs.

GitHub: GitHub was used to store and organize the source code of the internship projects and maintain the project repository.

5.7 System Architecture

The developed applications follow a three-layer architecture consisting of the Presentation Layer, Application Layer, and Output Layer. The architecture separates user interaction from program processing and result generation, making the applications easier to understand and manage.

1. Presentation Layer

This layer represents the graphical interface through which the user interacts with the application. It contains elements such as buttons, labels, input fields, selection controls, and display areas created using Tkinter.

2. Application Layer

This layer contains the main Python programming logic. It receives the user input and performs the required operations using functions, conditions, operators, lists, strings, randomization, and Python modules.

3. Output Layer

This layer displays the processed information to the user through the graphical interface. The output varies according to the application, such as the winner/result of a Rock Paper Scissors round, a generated password, or a calculated result.

User / Input
Presentation Layer Tkinter GUI
Application Layer Python Logic • Conditions • Functions • Randomization
Output Layer Game Result • Generated Password • Calculation Result

5.8 Methodology

Stage	Description
Requirement Analysis	The requirements and expected functionality of each project were studied before development.
Planning	The main features, application structure, game rules, and required interface components were planned.
Implementation	Python code was written to implement the required functionality of each application.
GUI Development	Tkinter was used to create the graphical interface and interactive components.
Testing	Different inputs, choices, passwords, and arithmetic operations were tested to verify that the applications worked correctly.
Debugging	Errors identified during testing were analyzed and corrected through code modifications.
Documentation	The completed projects and source codes were organized and maintained in the GitHub repository.

Project Workflow

The overall workflow followed a simple development cycle: understand the task requirements → plan the application → implement Python logic → create the GUI → test different cases → debug errors → organize and document the final project.

5.9 Expected Outcomes

The projects are expected to provide practical understanding of Python programming and application development.

The major expected outcomes include:

- Better understanding of Python programming fundamentals.
- Practical experience in developing GUI applications.
- Improved logical and problem-solving abilities.
- Understanding of functions, operators, lists, strings, conditions, and modules.
- Practical use of randomization in application logic.
- Experience in handling user input and application events.
- Improved debugging and testing skills.
- Basic experience with GitHub and project organization.

5.10 Certificates of Completion and Internship Proof

This section presents the supporting documents related to participation and successful completion of the Python Programming Internship. The Internship Completion Certificate issued by CODSOFT is included as official proof of completion. The Internship Offer Letter is also included as supporting proof of selection, internship duration, and the assigned Python Programming internship position.

The certificate confirms completion of four weeks of the virtual internship from 05 August 2026 to 05 September 2026. The offer letter states the one-month internship period beginning on 05 August 2026.

INTERNSHIP LEARNING SUMMARY

The internship provided an opportunity to apply Python programming concepts through small, functional applications rather than only studying concepts theoretically. The three projects covered different types of programming problems and helped connect logic with user interaction.

Rock Paper Scissors focused on conditional logic, user input, random computer choices, and game-result evaluation.

Calculator focused on arithmetic operations, event handling, input processing, and displaying calculated results through a graphical interface.

Password Generator focused on random character selection, string handling, user-defined length, and generation of stronger password combinations.

Overall, the internship strengthened practical programming, problem-solving, testing, debugging, and basic project organization skills. It also provided exposure to GUI development using Tkinter and source-code management using GitHub.

Internship Details

Organization	CODSOFT
Internship	Python Programming
Duration	05 August 2026 – 05 September 2026
Projects	Rock Paper Scissors, Calculator, Password Generator
Completion Date on Certificate	08 September 2026

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